

MARIAGIULIA MARTINELLI

Bioarchaeology | Palaeogenetics | Conservation Science

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PROFILE

I study what happens to biological material after deposition: the taphonomy of human skeletal remains, the microbial biodeterioration of stone, and the survival of ancient biomolecules. I work across **bioarchaeology**, **palaeogenetics** and **conservation science**, from osteological and laboratory practice through to reproducible, openly released R pipelines.

EDUCATION

Master's Degree in *Science and Technology for the Conservation of Cultural Heritage* (LM-11)

Sapienza University of Rome

📅 09/2024 – graduating 21/10/2026

- Thesis title: *Conservation and Reassessment of an Overlooked Skeletal Collection: Historical and Bioarchaeological Study of Human Crania from Etruscan Territory Acquired by G. Sergi in the Early 20th Century.*
- Thesis supervisors: Dr. Ileana Micarelli, Prof. Mary Anne Tafuri, Prof. Laura Maria Michetti
- Programme delivered in Italian and English; admission conditional on assessed B2 English (CEFR).
- Student Representative for the Department of Environmental Biology (Bachelor, Master and PhD level) and for the LM-11 Master's Degree Class.

Erasmus+ Exchange Student

Stockholm University, Sweden

📅 01/2026 – 06/2026

- Courses: *Palaeogenetics* (BL7069); *Modelling Tools for Environmental Scientific Studies* (MI4011); *Communicating Environmental Science* (MI4008).

Bachelor's Degree in *Tecnologie per la Conservazione e il Restauro dei Beni Culturali* (L-43)

Final grade: 110/110

Sapienza University of Rome

📅 09/2021 – 12/2024

- Thesis title: *Studio dell'evoluzione del microbiota sulle pareti del Casino Algardi (Roma) in seguito ad interventi conservativi con prodotti sostenibili.*
- Thesis supervisors: Prof. Enea Gino Di Domenico, Dr. Chiara Alisi, Prof. Viviana Fonti, Prof. Claudio Chimenti
- Student Representative of the L-43 Degree Class.

RESEARCH EXPERIENCE

Research Internship at the *Centre for Palaeogenetics* (CPG)

Stockholm University / Sapienza University of Rome

📅 04/2026 – 06/2026

- Analysed 250 ancient DNA samples from Roman, Visigothic, Islamic and Medieval necropolises in Valencia (Spain), sampled from petrous bone, teeth and phalanges.
- Ran Kraken2 (GTDB) and KrakenUniq (MicrobialNT) classification pipelines as SLURM job arrays on the Dardel HPC cluster (PDC, KTH). Assessed cross-tool agreement (Procrustes, Spearman, PCoA on Bray–Curtis) and screened all samples for *Yersinia pestis*, *Mycobacterium tuberculosis* and *Tannerella forsythia*.

Osteological Laboratory, *Tusculum* Project

EEHAR-CSIC (Escuela Española de Historia y Arqueología en Roma)

📅 12/2025 – 01/2026

- Catalogued and analysed medieval human remains from the Area G cathedral site (DG-ABAP no. 245): skeletal inventory and element identification, biological profile, and macroscopic recording of preservation, weathering and taphonomic alteration.
- Managed the digitisation of the anthropological dataset, ensuring traceability between remains, records and photographic documentation.

Research Internship at the *Palaeoanthropology and Bioarchaeology Laboratory*

Sapienza University of Rome

📅 07/2025 – 12/2025

- Bioanthropological study and critical re-evaluation of a human cranial collection acquired by Giuseppe Sergi (early 20th century): biological profile, dental palaeopathology, history of previous study, and systematic assessment of the state of conservation, distinguishing taphonomic damage from damage introduced during excavation and curation. The objective was to revitalise a neglected scientific heritage collection and propose new avenues for research.
- Designed the reproducible R pipeline used for the collection (*Etruscan-Analysis*).

Research Internship at the ENEA Research Centre, Casaccia (Rome)

ENEA Casaccia

📅 04/2024 – 12/2024

- Designed and executed independently an end-to-end metagenomics workflow on microbially mediated deterioration of a mineral substrate: field sampling and photographic documentation of biofilms on the historic walls of the *Casino del Bel Respiro*, genomic DNA extraction, PCR (16S rRNA, ITS), amplicon sequencing and phylogenetic classification.
- Built the R pipeline from scratch (phyloseq, vegan, ggplot2): alpha- and beta-diversity, taxonomic composition and functional-group succession under sustainable conservation treatments.

Research Internship: Taxonomic Identification and Iconographic Analysis

Sapienza University of Rome

📅 2022 – 2024

- Study of *The Four Elements* cycle by Jan Brueghel the Elder through taxonomic classification of the species represented.

AWARDS AND RECOGNITIONS

Tutoring Fellowship (Cat. B1), Degree Programmes in *Scienze Applicate ai Beni Culturali*

Department of Environmental Biology, *Sapienza University of Rome*

📅 a.y. 2026/2027

- Awarded through public comparative selection (Call BT-B1 18/2026).
- 75 hours of teaching and laboratory support for the L-43 and LM-11 degree classes (supervision: Prof. M. Beccaccioli, Prof. L. Medeghini, Prof. A. Ciccola).

Merit Scholarship – Academic Excellence Track (Master's)

Sapienza University of Rome

📅 03/2026 – 08/2026

- Highly selective merit-based scholarship and honours programme (**Percorso d'Eccellenza**) for top-ranked students, requiring an integrative curriculum of advanced research beyond standard degree requirements.
- Ancient metagenomic analysis of Roman-Medieval archaeological samples, at the *Centre for Palaeogenetics*, Stockholm.

Merit Scholarship – Academic Excellence Track (Bachelor's)

Sapienza University of Rome

📅 04/2023 – 12/2024

- Highly selective merit-based scholarship and honours programme (**Percorso d'Eccellenza**) for top-ranked students, requiring an integrative curriculum of advanced research beyond standard degree requirements.
- Iconographic and taxonomic analysis of the *Four Elements* cycle by Jan Brueghel the Elder.

Student Assistantship, Earth Science Library

Sapienza University of Rome

📅 2023 – 2025

- Merit-based student collaboration grant awarded through competitive University ranking.

PUBLICATIONS



C. Alisi, V. Fonti, M. Martinelli, G. Migliore, P. Paganin, F. Tasso

Abstract in the Proceedings of *The Omics & Heritage (O&H)*, Rome, Italy, 14–15 May 2024

Unveiling the Microbial Hazard: Exploring Biofilm Composition on Casino Algardi for Conservation Insights

MANUSCRIPTS IN PREPARATION



M. Martinelli, V. Fonti, P. Paganin, F. Tasso, C. Alisi

International Biodeterioration & Biodegradation (manuscript in preparation, 2026)

Microbial Biodiversity on the Casino del Bel Respiro Walls: Implications for Conservation Strategies



L. Morucci, A.A. Bastanza, M. Martinelli, S. Iorio

International Journal of Paleopathology (manuscript in preparation, 2026)

A Systematic Review of Analytical Methodologies in Bioarchaeology of Stress in Non-Adult Populations: Recommendations for the Next 50 Years



M. Martinelli, C. Chimenti

Manuscript in preparation, 2026

The Animal World in Jan Brueghel's Four Elements

ORAL CONTRIBUTIONS



Orizzonti Antichi tra Ordine Perduto e Nuove Armonie

Oral contribution *Glycyrrhiza glabra* vs Biofilm: nuove prospettive per la conservazione dei beni culturali

M. Martinelli, V. Fonti, F. Tasso, P. Paganin, C. Alisi; University of Pisa · 02–04/12/2025



The Omics & Heritage (O&H)

Co-author of oral contribution *Unveiling the Microbial Hazard: Exploring Biofilm Composition on Casino Algardi for Conservation Insights*

C. Alisi, V. Fonti, M. Martinelli, G. Migliore, P. Paganin, F. Tasso; CNR, ISB · 14–15/05/2024

OPEN SOURCE PROJECTS



Etruscan-Analysis: Bioarchaeological Pipeline (R),
to be released



CO₂ Time Machine (JavaScript)



Bioarchaeological-Systematic-Review (R)



scienze-abc (HTML)



kraken2-krakenuniq-comparison (R, Shell)



Biofilm Metagenomics Workflow (Casino Algardi),
to be released

SKILLS

-  **Osteology, Taphonomy & Conservation of Skeletal Remains**
Skeletal inventory and element identification · biological profile (age, sex, stature) · dental palaeopathology · macroscopic recording of preservation, weathering and taphonomic alteration · selection of elements and sampling loci · collection management, photographic documentation and record-specimen traceability
-  **Laboratory & Analytical Practice**
Genomic and ancient DNA extraction · PCR (16S rRNA, ITS) · amplicon sequencing · sterile technique and contamination control in low-copy-number contexts · optical and stereomicroscopy of heritage and biological materials · biofilm sampling
-  **Data Analysis, Statistics & Reporting**
R (tidyverse, phyloseq, vegan, ggplot2, sf, leaflet) · non-parametric testing · multivariate ordination (PCoA, Bray-Curtis, Procrustes) · reproducible pipelines, Git/GitHub, $\LaTeX$ · publication-quality figures, reports and manuscripts
-  **Bioinformatics & High-Performance Computing**
Kraken2 · KrakenUniq · read-count-based pathogen screening · SLURM job arrays on HPC (Dardel, PDC/KTH) · shell scripting · Linux command line
-  **Core Scientific Background**
General, organic and analytical chemistry · human anatomy and skeletal biology · microbiology and biology of deterioration · biostatistics
-  **Communication, Organisation & Autonomy**
Independent project management from sampling design to written report · elected student representative across three degree levels (L-43, LM-11, PhD) · conference presentations and co-authored manuscripts

LANGUAGES

-  **Italian**
Native speaker (C2). Full academic and professional use.
-  **English**
B2 (CEFR), assessed at admission to the LM-11 Master's programme, which is delivered in Italian and English. Full Erasmus+ semester at Stockholm University.

CONFERENCES, WORKSHOPS & WEBINARS

-  **International Conference on Palaeogenomics (ICP2026)**
Stockholm, Sweden · 23–26/06/2026
-  **Webinar: *Bottleneck o Consanguineità: come si entra in Accademia***
GIGA (Gruppo Italiano Giovani Antropolog3) · 15/12/2025
-  **Webinar: *Starter Pack per la ricerca: dalla scrittura della tesi ai primi congressi***
GIGA (Gruppo Italiano Giovani Antropolog3) · 26/11/2025
-  **Seminar: *Nuove opportunità professionali per i biologi nel mondo dell'arte e dei beni culturali***
FNOB, FIB · 27/02/2025
-  **Webinar: *Primi passi con R: introduzione al linguaggio per analisi statistiche***
Consortium GARR · 12/06/2024
-  **SAGacia24: *Primo Simposio Arte che Guarda alla Scienza***
Sapienza University of Rome · 05/04/2024
-  **Workshop: *Dagli Acetali, ai DES, ai FAME: i nuovi solventi green introdotti da YOCOCU nel settore dei Beni Culturali***
YOCOCU · 29/01/2024
-  **Seminar: *La meteorologia moderna e il ruolo guida dell'Italia nel suo sviluppo storico***
CREA, Università della Tuscia · 29/11/2023

WORK EXPERIENCE

Library Assistant
Library of the Department of Earth Sciences, *Sapienza University of Rome*
📅 01/2023 – 01/2025

Private Science Tutor
Self-employed
📅 2021 – 2023

Tennis Instructor
Reschio Hotel Summer Camp
📅 06/2021 – 08/2021

CERTIFICATIONS

-  **Open Badge: Student Representation**
Issued by Bestr · 2025
-  **Open Badge: Service Learning**
Issued by Bestr · 2024
-  **Course: D.Lgs. 81/2008 Health and Safety Training**
KPMG Advisory · 27/02/2023
-  **Certificate of Excellence: Localizing the Sustainable Development Goals (MUSTER)**
Issued by Dr. Karl-Ludwig Schibel, President of the Fiera delle Utopie Concrete, Liceo Scientifico “Plinio il Giovane” · 2018

REFERENCES

- Dr. Ileana Micarelli**
@ *Sapienza University of Rome*
✉

Dr. Chiara Alisi
@ *ENEA Casaccia*
✉
- Dr. Gonzalo Oteo**
@ *Centre for Palaeogenetics, Stockholm University*
✉

Prof. Enea Gino Di Domenico
@ *Istituti Fisioterapici Ospitalieri*
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Autorizzo il trattamento dei dati personali contenuti nel mio curriculum vitae ai sensi del Regolamento UE 2016/679 (GDPR) e della normativa italiana vigente. Acconsento inoltre a che la presente candidatura sia presa in considerazione nei 12 mesi successivi alla selezione anche per ulteriori posizioni con profilo professionale simile.

Signature: *Mariagiulia Martinelli*

Date: August 31, 2026